Chapter 12 Problem Set

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1. There are $\binom{20}{7}$ ways to choose which cubes are blue, $\binom{13}{8}$ to choose which are green, $\binom{5}{2}$ for red, and $\binom{3}{3}$ for yellow. Multiplying them together, we get $77520 \cdot 1287 \cdot 10 \cdot 1 = 997682400$.

2a. Finding the number of subsets of [10] that contain 1 is equivalent to finding the number of subsets of $\{2, 3, \ldots, 10\}$, which is $2^9 = 512$.

2b. To do this, we can find the number of subsets that contain 1 but not 2, the number of subsets that contain 2 but not 1, and the number of subsets that contain both 1 and 2, and add all three numbers. The number of subsets that contain 1 but not 2 is 2^8, and the same for the number of subsets that contain 2 but not 1, and the subsets that contain both 1 and 2. Therefore, the answer is $3 \cdot 2^8 = 768$.

2c. We know that there are 2^{10} subsets in total, so we can subtract from that the number of subsets containing only even numbers, which is 2^5; $2^{10} - 2^5 = 992$.

3a. The function $f : Z \to Z, f(n) = n$ is surjective and injective because f maps every integer to the same integer. Therefore, it is also bijective.

3b. The function $f : Z \to Z, f(n) = n^2$ is not surjective because every non-perfect square doesn't have a corresponding input value. It is not injective because an integer and the negation of that integer have the same square.

3c. This one is the same as the previous one, except it is injective because no two distinct positive integers have the same square.

3d. The function $f : Z \to Z, f(n) = \lfloor \frac{n}{3} \rfloor$ is surjective because every integer is equal to the floor of three times that integer over 3. However, it is not injective because, for example, $\lfloor \frac{3}{3} \rfloor = \lfloor \frac{4}{3} \rfloor$.

4. We must choose at least one element from one of the S_i's. Assume that we choose S_1 for that. Then, from every other set, we can either choose none or one of its elements. Therefore, if n_i is the number of elements in that set, then there are $n_i + 1$ ways to deal with it. So, assuming that we must choose an element

from S_1, there are $n_1(n_2+1)(n_3+1)\cdots(n_k+1)$ ways to satisfy the conditions of the problem. Doing the same for S_2, S_3, through S_k, we see that the total sum is $n_1(n_2+1)\cdots(n_k+1)+n_2(n_1+1)(n_3+1)\cdots(n_k+1)+\cdots+n_k(n_1+1)\cdots(n_{k-1}+1)$.

5. Without loss of generality, let's assume that m is the number of rows. The k rooks must be on k different rows, so we need to choose k rows from m total. This gives us $\binom{m}{k}$ ways to choose the rows that the rooks will be placed on.

Now, we must choose which of the n columns will have rooks on them. Since there are k rooks, this is just $\binom{n}{k}$. Now, when we multiply that by the number of ways to choose the k rows, we get $\binom{m}{k}\binom{n}{k}$ ways to place k rooks on an $m \times n$ chessboard.

6a. Let's say f maps $[m]$ to $[n]$. Each input value from 1 to m has n possible output values. Therefore, there are m^n ways to map $[m]$ to $[n]$.

6b. Now that f is injective, once an output value is taken, f can't map any other input value to that output. That means each output value is mapped to by one input value. There are m choices for the first output value, $m-1$ for the next output value, and so on until either we run out of input or output values. If $m > n$, this gives us $n!$ possible functions. If $m < n$, there are $(n-m)!$ possible functions.

7. The number of compositions of a number n is 2^{n-1}. We know that n can be written as $1+1+\ldots+1$ with n ones. To generate a new composition, we can either add two of the ones connected by the plus sign between them to create a new part, or we can leave them as individual parts. Let these actions be represented by the numbers 1 and 0 respectively. Then, we can create a set S with $n-1$ elements, which are either 0 or 1.

Now, we need to prove a bijection between the compositions of n and the number possible sets S. We know that such a relation is surjective because, given a binary set of length $n-1$, we can construct a composition of n by interpreting each 1 as creating a new part and each 0 as keeping the current part. So we can create a composition of n given any set S.

We know that it is injective as well, because given a composition of n, we can construct a binary set using 1 to split a part of the composition, n_i, into 1 and $n_i - 1$, until we get a sum of $n-1$ ones.

Therefore, there is a bijection between the number of compositions of n and the number of possible sets S. Since there are $n-1$ terms which can each be 0 or 1, there are 2^{n-1} possible sets S, and therefore 2^{n-1} compositions of n.

8. Let's consider all compositions of n with k parts. From the problem 9,

we know that there are $\binom{n-1}{k-1}$ of them. When $k = n$, the only composition is $1 + 1 + \ldots + 1$ with n ones. That composition has zero even parts.

When $k = n - 1$, the composition consists of $n - 2$ ones and a two. That means it has an odd number of even parts (1). When $k = n - 2$, either the 2 from the previous compositions is turned into a 3 or two of the ones combine to form a 2. Either way, there is an even number of even parts in the new composition.

This pattern continues until $k = 1$; when $k = n - a$ where a is even, then all such compositions have an even amount of even parts. When a is odd, there is an odd number of even parts. Therefore, our overall sum becomes $\binom{n-1}{k-1}+\binom{n-1}{k-3}+\ldots+\binom{n-1}{n-2\lfloor\frac{n-1}{2}\rfloor}$. Expanding $(x+y)^{n-1}$ and $(x-y)^{n-1}$ using the Binomial Theorem, subtracting the second equation from the first, and substituting $x = y = 1$ shows us that the above expression is equal to 2^{n-2}. Therefore, there are 2^{n-2} compositions of n with an even number of even parts.

9. We saw that we can use a set of $n - 1$ zeros and ones to represent any composition. If a composition has k parts, then that set needs to contain $n-(k-1)$ ones. So there are $\binom{n-1}{n-(k-1)}=\binom{n-1}{k-1}$ compositions of n with k elements.

11. We can fill the first $m - 1$ rows and $n - 1$ columns however we want, and the rest of the board depends on what we choose. Therefore, the answer is $2^{(m-1)(n-1)}$.